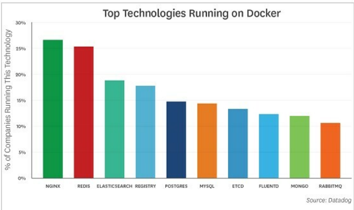

Figure 12: Top technologies/companies running on Docker technology

Figure 13: Docker hosts often run seven containers at a time

2. *Redis:* This popular key-value data store is often used as an in-memory database, message queue, or cache.
3. *ElasticSearch:* Full-text search continues to increase in popularity, cracking the top three for the first time.
4. *Registry:* Eighteen per cent of companies running Docker are using Registry, an application for storing and distributing other Docker images. Registry has been near the top of the list in each edition of this report.
5. *Postgres:* The increasingly popular open source relational database edges out MySQL for the first time in this ranking.
6. *MySQL:* The most widely used open source database in the world continues to find use in Docker infrastructure. Adding the MySQL and Postgres numbers, it appears that using Docker to run relational databases is surprisingly common.
7. *etcd:* The distributed key-value store is used to provide consistent configuration across a Docker cluster.
8. *Fluentd:* This open source 'unified logging layer' is designed to decouple data sources from backend data stores. This is the first time Fluentd has appeared on the list, displacing Logsout from the top 10.
9. *MongoDB:* This is a widely-used NoSQL datastore.
10. *RabbitMQ:* This open source message broker finds plenty of use in Docker environments.

vii) *Docker hosts often run seven containers at a time*

The median company that adopts Docker runs seven containers simultaneously on each host, up from five containers nine months ago. This finding seems to indicate that Docker is in fact commonly used as a lightweight

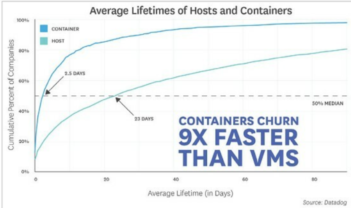

Figure 14: Containers' churn rate is 9x faster than VMs

way to share compute resources; it is not solely valued for providing a knowable, versioned runtime environment. Bolstering this observation, 25 per cent of companies run an average of 14+ containers simultaneously.

viii) *Containers' churn rate is 9x faster than VMs*

At companies that adopt Docker, containers have an average lifespan of 2.5 days, while across all companies, traditional and cloud-based VMs have an average lifespan of 23 days. Container orchestration appears to have a strong effect on container lifetimes, as the automated starting and stopping of containers leads to a higher churn rate. In organisations running Docker with an orchestrator, the typical lifetime of a container is less than one day. At organisations that run Docker without orchestration, the average container exists for 5.5 days.

Containers' short lifetimes and increased density have significant implications for infrastructure monitoring. They represent an order-of-magnitude increase in the number of things that need to be individually monitored. Monitoring solutions that are host-centric, rather than role-centric, quickly become unusable. We thus expect Docker to continue to drive the sea change in monitoring practices that the cloud began several years ago. END

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